

# The Evolution of Knowledge in the Refactoring Research Field

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## ABSTRACT

Refactoring is certainly one of the most widespread practices used by developers to improve software quality. During the last two decades, it has been the subject of an increasing number of research studies. But how much do we know about the trends, and the appearance of new topics, in the research field of refactoring? What have researchers found about its application since the time of its first introduction? In this work we provide a preliminary analysis of the state of the art and the evolution of the research on refactoring. We attempt to represent the actual body of knowledge in this field through the analysis of its cognitive structure, leveraging science mapping methodology to focus on the most relevant concepts in this research area. We model the body of knowledge by mining bibliographic databases and by retrieving the co-occurrence of keywords. We have found that some different general themes can be recognized, but not all of them have the same role and equally catalyzed researchers' interest. In addition, we provide a preliminary analysis on the trends and directions as well as the hot topics we identified in the refactoring research field.

## CCS Concepts

• **Software and its engineering** → **Agile software development**; **Object oriented development**; *Software design engineering*;

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## Keywords

Refactoring, Science Mapping, Bibliographic Network

## 1. INTRODUCTION

It has been 15 years since the first edition of Fowler's work [10] and even more since the time of Opdyke's PhD dissertation [15]. It is time for researchers involved in software engineering to ask what has been done so far and which are the future perspectives in the research field of refactoring. Different authors have recently focused their attention on analyzing the research on Refactoring [1, 13, 2, 20]. This suggests that researchers show interest about the current state and future directions of the refactoring research field. However, with the exception of [1], most of the previous works addressed specific aspects (e.g., bad smells) and did not focus on the entire scientific production on Refactoring. This preliminary work tackles the problem of representing the up-to-date body of knowledge using a different approach, based on science mapping [5].

Science mapping gives researchers a mean for representing the existing relations among research themes within a research field. Science mapping allows to discover research sub-areas, and returns information on the cognitive, intellectual and social structure as well as the evolution of a scientific discipline. In this work, we use science mapping tools to study the cognitive structure of the research on refactoring. We aim at revealing existing relationships among the main themes investigated by researchers so far, and their evolution through time. The best model to represent the properties we are interested in is the so called *co-occurrence network*. In a co-occurrence network, nodes are the keywords found in publications corpus, and edges represent co-occurrence of two keywords within the same publication.

We followed the process illustrated by Cobo et al. [5]. First, we retrieved bibliographic data by mining the ISI Web of Science (ISIWoS) database [14], one of the most important bibliographic database. Second, we discarded useless or redundant data through a pre-processing stage. Then, we

built the co-occurrence network and finally we performed a complete science mapping analysis on the retrieved network. The obtained results allowed us to identify main and side research themes on software refactoring, reveal research trends, and recognize abandoned topics. Our study could also help researchers in understanding how their specific interests fit in the wider context of the research on Refactoring, or how the topic they are most interested in evolved over time. A similar approach, related to a different research field, was already presented in [6] and [16]. However, no similar studies had been conducted about Refactoring. In the refactoring research field the most comprehensive works are that of Mens et al. [12] and the more recent work of Abebe et al. [1]. However, in the mentioned studies the authors used a totally different approach, and did not consider bibliographic networks.

This work is structured as follows. Section 2 reports the background knowledge useful for understanding the following sections. In Section 3 a thorough description of the methodology adopted in this work is reported, whereas in Section 4 we give a detailed description of how we set our experiment. The next Section 5 reports the results we obtained. Finally some threats to validity are illustrated in Section 6 and then in Section 7 we draw some conclusions, also outlining some extensions of this work, which we are considering for the next future.

## 2. BACKGROUND

Science Mapping provides a spatial representation of a discipline, highlighting the reciprocal relationships among its elements [19]. This way it is possible to study the structure and evolution of a discipline. In order to achieve this goal it is possible to use several techniques (co-citation analysis, co-word analysis, etc). We applied this mapping to the research field of Refactoring in software engineering, performing a co-word analysis on the keywords used by authors to characterize their papers. Introduced by Callon [4], co-word analysis focuses on terms contained in documents, analysing how these terms are shared among different elements of a body of knowledge.

Co-occurrence analysis has been used in several research fields, specifically of technical area, including software engineering [9]. A work dealing with a similar problem using the same approach, but applied to a different research field (chemical engineering), is that of Peters et. al [16]. In the field of software engineering there is only the study of Coulter [9], that analyzed the research in software engineering using a keyword co-occurrence analysis. In the present work we focus on a specific research niche, software Refactoring, and we use a wider set of methods and tools. In the Refactoring field there has been recently some attempts to summarize the results collected during the last years. By and large the authors of these works focus on specific issues (refactoring opportunities [2], code smells [20], etc.) and did not focus on the entire literature on Refactoring. For example Al Dallal et al. [2] carried out a research focused on methods to detect refactoring opportunities, whereas in Vale et al [20] the authors studied the literature on Refactoring with specific regards to the code smell they fix. On the other hand, to the best of the authors' knowledge, systematic surveys about Refactoring were carried out by Mens et al. [12] and Abebe et al. [1]. In the first work the authors studied the available literature on techniques and formalism,

types of software artifacts, effect of Refactoring on software process and Refactoring activities whereas in the second one the authors focused their attention on trends and opportunities. None of these works used the approach proposed in the present paper.

In this work we attempt to study the cognitive structure of the software refactoring research field, and how this structure evolved during the a period from 1999 to 2015. For perform this analysis we rely on a software called SciMat [8].

## 3. METHODOLOGY

For the science mapping analysis we adopted the workflow presented in [5], that we briefly outline in the following paragraph.

### 3.1 Science Mapping Process

A science mapping process is basically structured in the following steps:

**Data Retrieval** The main source for our research is the ISIWoS bibliographic database [14]. We queried the ISIWoS database to retrieve works dealing with software Refactoring.

**Preprocessing** In order to avoid misleading results, a preliminary preprocessing phase is required. This implies to look for duplicated records, to remove plurals, and so on. To obtain relevant information we discarded redundant or useless information. For example, we decided to discard all-comprehensive keywords, like "refactoring" (all papers deal with it), "software", and similar keywords.

**Network Extraction** In this step we built a co-occurrence network, namely a network where nodes are keywords and edges represent a co-occurrence of two keywords in the same paper. Edges can also be weighted by the number of papers where the two keywords co-occurred. In this preliminary analysis we set all weights to one.

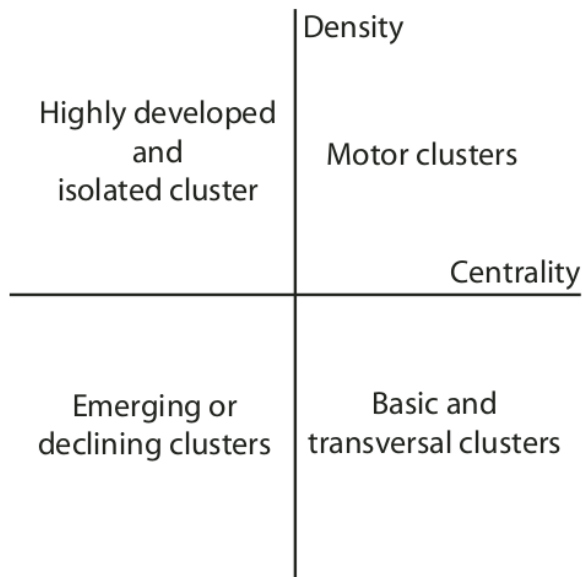
**Normalization** At this stage of the science mapping process, it is required to choose a similarity measure which is going to be used in the next stage for clustering purposes. A similarity measure determines how much two elements are similar. As a normalization metric we used the Jaccard Index.

**Mapping** In this phase a clustering algorithm is applied, with the aim of understanding how and to which extent the bibliographic network is structured in subnetworks, namely to identify groups of densely connected elements or clusters. Afterwards, a mapping process is needed in order to associate documents to the clusters they belong to.

**Analysis** Several analysis could be performed at this step. In our case we are interested in a co-word analysis.

**Visualization** In this work we represent our data through a strategic diagram.

**Interpretation** Based on the knowledge of the field, it is possible to draw some conclusions about the structure and evolution of the research field on Refactoring.



**Figure 1: Cluster typologies in a strategic diagram (from [8])**

### 3.2 Bibliographic databases

There are several bibliographic databases: ISIWoS [14], Scopus [18], PubMed [17], GoogleScholar [11], just to name a few of the most popular, having each of them its own peculiarity. In this work we decided to use data retrieved from ISIWoS. We chose ISIWoS over other databases for several reasons. In the first place, ISIWoS is widely considered one of the most selective bibliographic databases, recognized world wide and accessible by all main research institutions or societies, such as universities and research centres; in the second place, it presents highly standardized data, and this allows for an easy replication of the study; moreover, ISIWoS is highly structured, and this results in quick and reliable queries; finally, we wanted to focus on works published on journals, because those publications are not influenced by deadlines and keywords in the call of papers as works submitted to conferences.

### 3.3 Co-occurrence Network

This work is specifically aimed at studying the conceptual structure and evolution along the time of the research field on Refactoring. This kind of studies might be performed by means of the co-occurrence network. A co-occurrence network (and the relative matrix) is composed by nodes that represent keywords associated to a paper by their authors, and edges connecting nodes represent the co-occurrence relationships among them. When two keywords  $i$  and  $j$  appear together in the same document, then a co-occurrence relationship between them is detected. It is also possible to associate the number of document where the keywords appear to the nodes and the number of document where they co-occur to the edges.

In this work we retrieved the co-occurrence adjacency matrix that can be associated to a corresponding co-occurrence network. After having retrieved the co-occurrence matrix, we computed the similarity among keywords by means of

an index, the *equivalence index* [3]:  $e_{i,j} = \frac{c_{ij}^2}{c_i c_j}$ , where  $c_{ij}$  is the number of documents in which the keywords  $i$  and  $j$  appear together, whereas  $c_i$  and  $c_j$  represent the number of documents in which the keywords  $i$  and  $j$  appear, respectively. If the two keywords occur together in each document, then the equivalence index is 1, whereas if they are never found together in the same document the equivalence index is 0. After computing the equivalence index it is possible to perform a clustering analysis, for example by means of the Simple Centre algorithm [9], in order to group keywords into thematic areas and establish associations between words. In this way it is possible to obtain the so-called "thematic networks". The name assigned to each thematic network corresponds to the most relevant keyword for the associated theme (usually, the most central keyword for the theme, or cluster).

### 3.4 Strategic Diagram

Once thematic networks between keywords are created, we can report the results on a *strategic diagram* like that portrayed in Fig. 1. A strategic diagram is a two dimensional diagram where thematic areas are reported based on their associated values of *centrality* (on the horizontal axes) and *density* (on the vertical axes). Centrality and density are two metrics that measure the interaction of bibliographic unit of some kind, not only keywords. Specifically, centrality measures how strong are the ties of a given theme with the other themes and is defined as:  $c = 10 \sum e_{kh}$ , where  $k$  ranges over words belonging to the network of the given theme, and  $h$  are words belonging to the other different thematic networks. It returns an information about how much a theme is relevant with respect to the others of the same research field and could be seen as a measure of external cohesion with other themes.

On the other hand density represents a measure of internal cohesion, since it measures how strong are the links among keywords belonging to the same theme. It is defined as it follows:  $d = 100 \sum \frac{c_{ij}}{n}$ , where  $i$  and  $j$  are two words belonging to the same theme and  $n$  the total number of the keywords for that specific theme. In some sense, the density measures how much the theme is a well developed topic. Depending on their reciprocal position in a strategic diagram, research themes could be classified as:

**Motor Themes** : characterized by both high centrality and density, they are well developed and relevant themes.

**Peripheral themes** : although well developed themes, they are more isolated if compared to the others. They usually correspond to highly specialized themes.

**Emerging or declining themes** : Having both low centrality and density, these themes are disappearing or emerging themes.

**Basic themes** : Not well developed but relevant themes. They are usually general themes.

Their respective position is shown in Fig. 1. Strategic themes are usually represented as spheres of different size, this way adding a third dimension to the diagram. Spheres' size is generally associated to a bibliometric measure of performance (e.g. average number of documents).

|    | Years             | N. Documents |
|----|-------------------|--------------|
| P1 | from 2001 to 2004 | 66           |
| P2 | from 2005 to 2009 | 151          |
| P3 | from 2010 to 2014 | 225          |

**Table 1: Number of documents per year from ISIWoS**

## 4. EXPERIMENTAL SETTINGS

The analysis we performed is structured in several steps. First and foremost we queried the selected bibliographic database, *ISIWoS*. In the query we used on *ISIWoS*, we searched for the word “refactoring” either if it appeared in the title, in the abstract, among the keywords or in the text. We considered only works written in English, selected among books, articles and reviews. After performing a query with such constraints, *ISIWoS* returned 432 works.

In order to avoid as much as possible the occurrence of false positive results, namely works that were retrieved by the search engine but are not actually related to software Refactoring, we manually checked each result. We found and discarded 16 out of the 432 works that are not actually related to the Refactoring topic (at least not in the sense intended in the software engineering field) and that ended up in the *ISIWoS* search results because the term “refactoring” was in the abstract, though it has a different meaning.

**Periods** We consider 3 periods, each of them of 5 years, starting from 2001 to 2014. Our query to *ISIWoS* did not return works for 1999, 2000 and 2015.

**Network** We retrieved the co-occurrence network, specifically considering only author’s keywords.

**Normalization** As a measure of normalization we decided to use the Jaccard Index [16].

**Clustering Algorithm** In order to retrieve information about how the nodes of the co-occurrence network are connected we need to compute which and how many clusters are structured in. We chose Simple Centre Algorithm as a clustering algorithm [9].

**Mapper** To map documents to each cluster we consider a Core Mapper. It returns the documents that are present in at least two nodes, and can be generalized to k-nodes [5].

**Performance** As a measure of performance we consider the average number of citations (see SciMAT user guide for more details [7]).

## 5. RESULTS

The first trend that can be inferred from this analysis is the regular increase in the number of research works on Refactoring, reported in Tab. 1, where the temporal distribution is shown. It must be noted that *ISIWoS* does not report works for the years 1999, 2000 and 2015. We performed an analysis with strategic diagrams for the three periods (years 2000-2004, 2005-2009 and 2010-2014), using both core-documents count and average-citations count. For each diagram the volume of the spheres is proportional to number of core documents, or to the average-citations count,

respectively. The diagrams show how the number of thematic networks increased along the three periods, from now on denoted as P1, P2, P3, from 3 thematic networks in P1, to 5 in P2, to 7 in P3 (Figs. 2, 3, 4).

Going from P1 to P3, thematic networks can change name. This is because SciMat labels each cluster using the most connected keyword within it. This could lead to two different cases. On the one hand, if the most connected keyword inside the cluster changes, the cluster name changes too. On the other hand, the most connected keyword can move from one cluster to another, and the cluster must change name. Thus, the labels in the strategic diagrams are mainly used for grouping many different subtopics belonging to the same cluster.

**P1** - The first diagram shows only three clusters. The cluster *Object-Oriented-System* is the main motor theme, followed by the cluster *Programming*. The cluster *Architecture* is isolated. The cluster *programming* has a much higher average-citation number.

**P2** - There are only two motor themes: *Agile-Methodologies* (Fig. 8) and *Languages*. *Agile-Methodologies* has the highest density, less documents than *Languages*, but more average-citations. It must be noted that these appear as new thematic networks in P2, but their keywords existed already in P1, and were two sub-topics of *Object-Oriented-System*. This means that the motor theme of P1 splits in more than two new motor themes, with different labels, in P2, but containing mostly the same topics, as we will discuss in more depth later on. Similarly to P1, the most cited thematic networks are in the centre of the diagram (i.e., *Reverse-Engineering* and *Object-Oriented-Systems*, visible in detail in Fig. 9). The cluster *Architecture* is still isolated, but their inner subtopics changed even if the label remained the same.

**P3** - A new main motor thematic network, *Experimentation*, appears. Also *Software-Evolution* and *Specific-Refactorings* are motor themes, even if to a less extent. Again, themes in the centre of the diagram collect more average citations than the motor theme.

**Trends.** The time evolution clearly shows how much dynamic the research field of Refactoring is. In all periods most documents are in the first and third quadrants, except for *Refactoring Classification*, whose network contains also *Refactoring Taxonomy*, shown in P3, meaning that the corresponding theme has become more specialized. This is probably due to the release of new automatic tools for detecting different kinds of refactorings, available since 2010 (e.g., RefFinder). Such tools provided an easy way for conducting empirical analysis on big amounts of data, recovering Refactoring classification and taxonomy from the various releases of a big Java project. All other clusters tend to align along the first and third quadrants, thus most of them represent emerging, declining or motor themes. The diagrams then reveal a dynamic behavior where subtopics move frequently from one thematic network to another.

*Comparisons: P1-P2.* The cluster labelled *Object-Oriented-Design* in P1 collects various relevant keywords. Considering the keyword *Software-Metrics*, the associated number of research works is even higher than that associated to *Object-Oriented-Design* (Fig. 5).

First of all, it must be noticed that three of the keywords that were members of the *Object-Oriented-Design* cluster in P1 are within cluster *Object-Oriented-System* in P2, namely, *Software-Metrics*, *Software-Maintenance*, and *Reuse*. Besides, *Object-Oriented-Design* is still coupled with keyword *Languages* in P2, since both of them are part of one of the two motor clusters, that is, the one labelled *Languages*. Moreover, other keywords that were previously located in the *Object-Oriented-Design* cluster are members of *Agile-Methodologies*, another motor thematic network in P2. Thus, together with the increase in the number of thematic networks in P2, the motor cluster *Object-Oriented-Design* in P1 splits into three new clusters, namely, *Agile-Methodologies*, *Languages*, and *Object-Oriented-Systems*, which also comprises new keywords. First, we can see how the keyword *Agile-Methodologies* acts as an aggregating node for the omonymous motor cluster, even if *Agile-Methodologies* itself does not appear in so many works as the other keywords of the same cluster do. Second, the keyword *Java* is gaining relevance inside the motor cluster labelled *Languages* (Fig. 8). In contrast to *Java*, *Smalltalk* is not to be found among the motor clusters in P2. Indeed, the couple *Object-Oriented-Design* and *Languages* still exists in a motor thematic network (i.e., *Languages*), but they are not linked to *Smalltalk* anymore. They are now linked to *Java*. This is not the sole relevant change: the keyword *Eclipse* is also associated to *Languages*. This clearly shows how Java and Eclipse popularity in the research field of Refactoring are the two sides of the same coin, as far as P2 is concerned.

*Comparisons: P2-P3.* It is immediately noticeable how the keyword *Agile-Methodologies* loses its aggregating role for the Refactoring motor themes. However, *Agile-Methodologies* stays in the first quadrant in P3, but with a more marginal role, even though it is still within a motor theme *Software-Evolution*. The two keywords *Model* and *Testing*, previously linked to agile methodologies in P2, are now grouped within *Specification-Refactoring*, a motor theme which also includes *Software-Metrics*, still associated with a great number of documents. In this period, the cluster *Experimentation* has the leading role as motor theme. *Languages*, as well as *Design*, is still within the most relevant motor keywords. It can also be noted that *Object-Oriented-Systems* moves from the first quadrant in P1 (inside *Programming* as motor theme as in Fig. 2) to the fourth quadrant in P2 (basic theme), to the third quadrant in P3 (emerging or declining term), suggesting a decreasing attention of researchers for this specific topic during the years.

## 6. THREATS TO VALIDITY

Our study suffers from some threats to validity that are reported below, according to the distinction among internal, external and construction validity.

**Internal validity** - We consider that the evolution of a discipline could be detected by analyzing the keywords chosen by the research for their papers. Despite the fact that the co-word analysis is a known methodology [4] and has been already used for the same kind

of study, we can not be sure that is sufficient to thoroughly represent the cognitive structure of the discipline. Perhaps it could be useful to address this issue also with other approaches that involve the analysis of the entire body of text.

**External validity** - For most of the computation we relied on SciMat [8]. Even if we consider this software reliable, we can not exclude that computation performed using different systems could lead to different results.

**Construction validity** - In order to retrieve our dataset we queried the bibliographic database *ISIWoS*. There exists other bibliographic databases that we did not yet addressed (Scopus, Google Scholar, etc). Although we retrieve a significant number of data from a reliable dataset we can not be sure that the analysis could not lead to different results with a different dataset retrieved from a different database. For instance, as this preliminary study does not focus on works published in conferences, it would be interesting to shift our attention to the potential existing connections between keywords and conferences in a future extension work. The present work could then serve as a solid ground to start from for a more specific research on this topic.

Finally, results could also be affected by a relative bias resulting from our experimental settings (e.g., the kind of algorithm chosen for clustering).

## 7. CONCLUSIONS

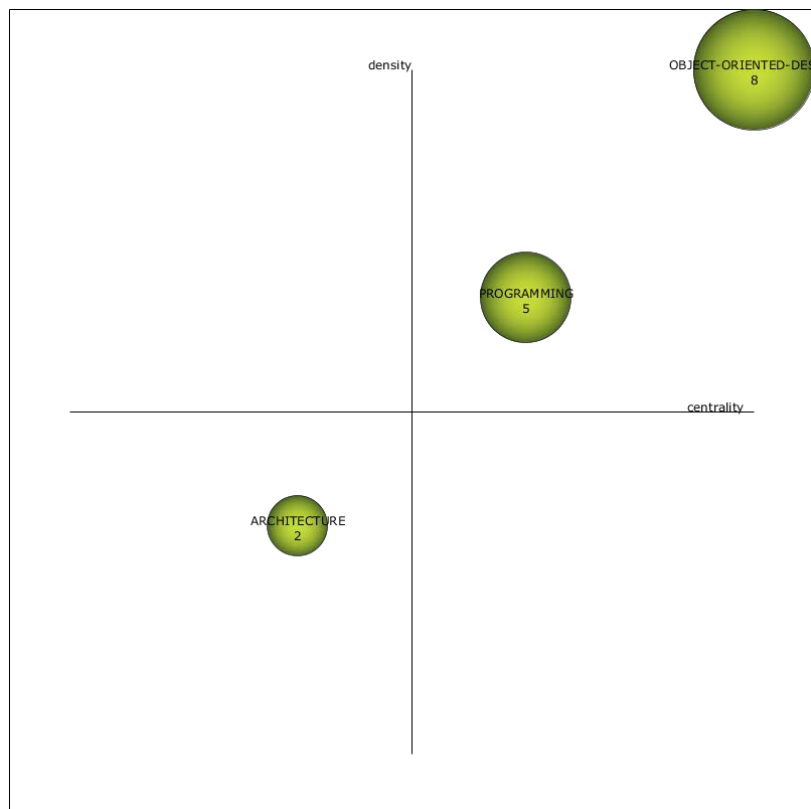
In this work we investigated the cognitive structure of the research on Refactoring and how it evolved along the time. After comparing the strategic diagrams for the three selected periods, we can conclude that agile methodologies had a leading, aggregating role in the period 2005-2009, even though this has not happened neither during the period 2001-2004 nor the period 2009-2015. Another relevant conclusion is that, during the same period in which agile methodologies took the lead in the evolution of the Refactoring research field, Java and Eclipse seemingly supported each other in the race for popularity among researchers. Finally, the last period analysed gives us an insight in the future directions of the evolution of the research field of Refactoring: highly specialized themes are appearing, and this is probably due to the increasing use of Refactoring tools not available in the past. Agile methodologies, even though not in the golden age of the years 2005-2009, are still present among the motor themes in the research field of Refactoring.

## 8. ACKNOWLEDGMENTS

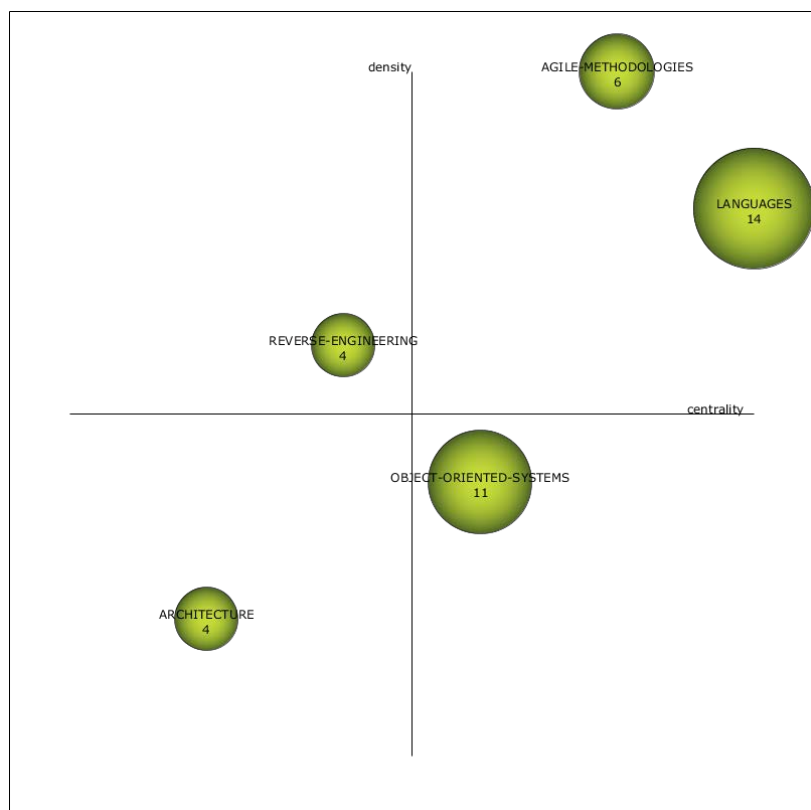
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**Figure 2: Strategic diagram for the period 2001-2004.**



**Figure 3: Strategic diagram for the period 2005-2009.**

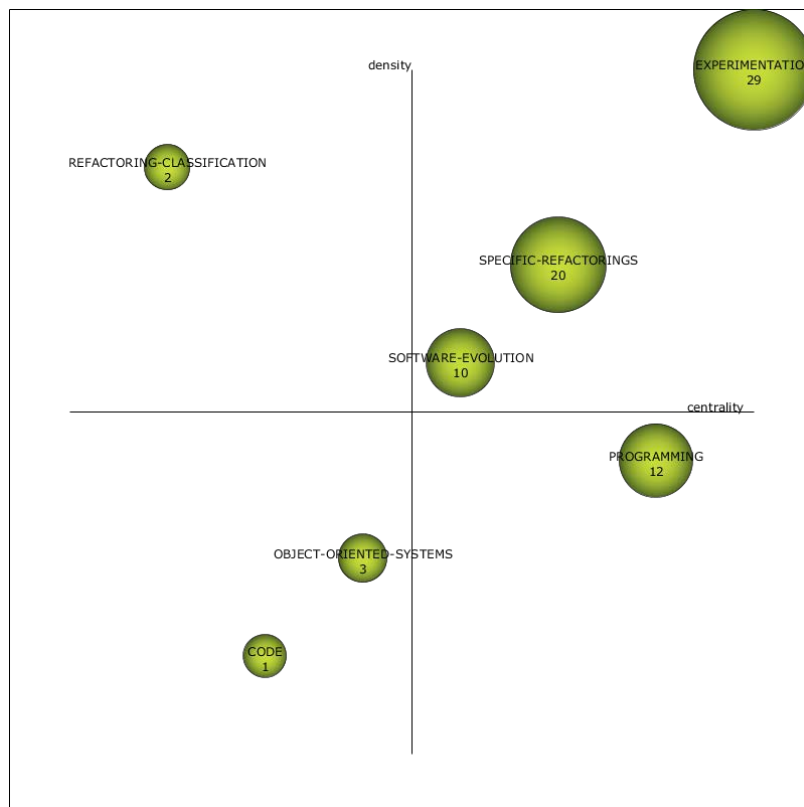


Figure 4: Strategic diagram for the period 2010-2014.

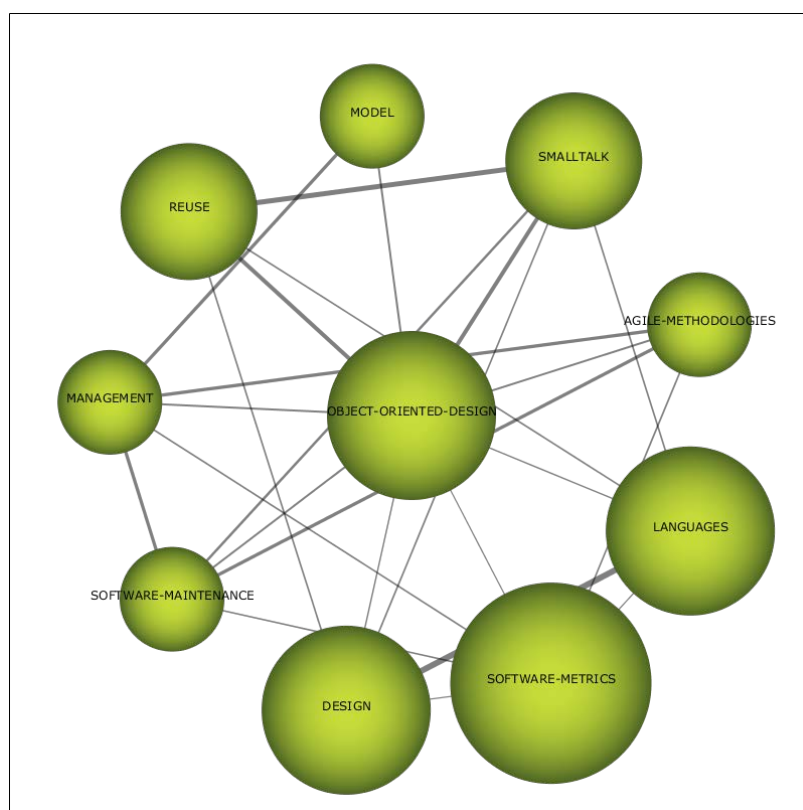


Figure 5: Thematic network *Object-Oriented-Design* - period 2001-2004.

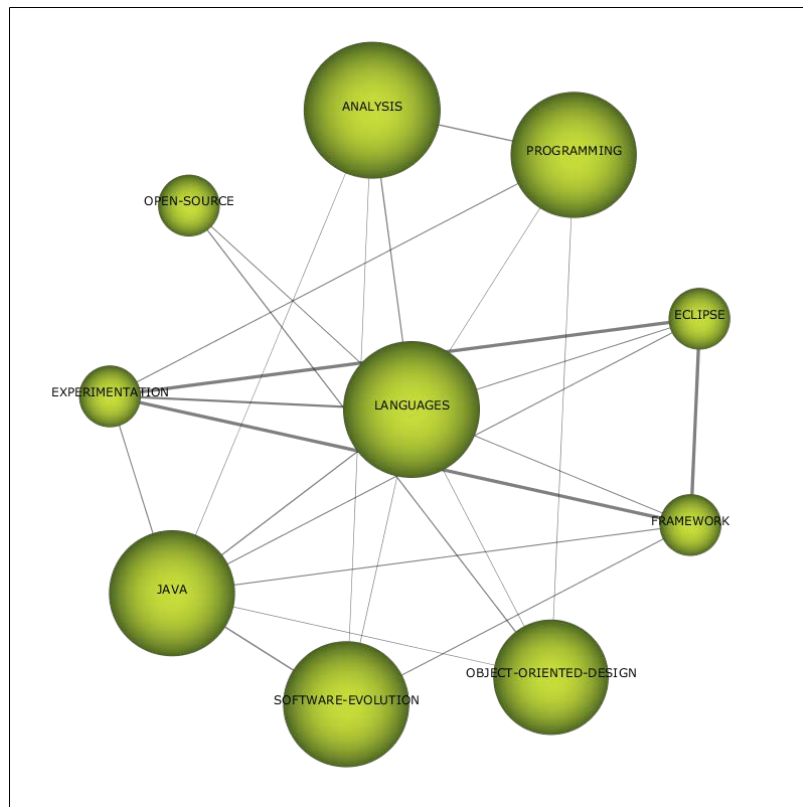


Figure 6: Thematic network *Languages* - period 2005-2009.

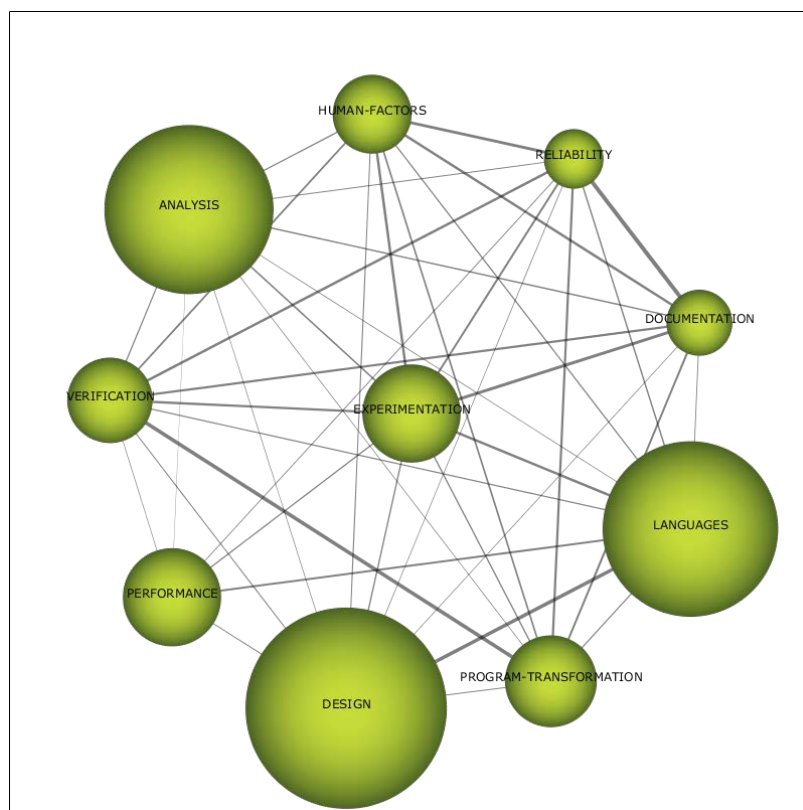


Figure 7: Thematic network *Experimentation* - period 2010-2014.

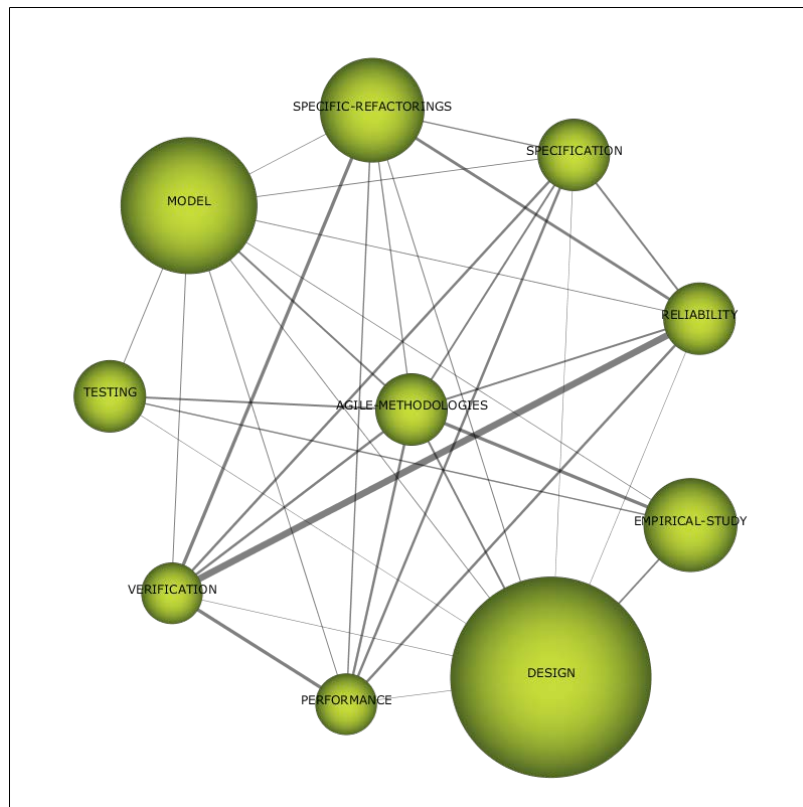


Figure 8: Thematic network *Agile-Methodologies* - period 2005-2009.

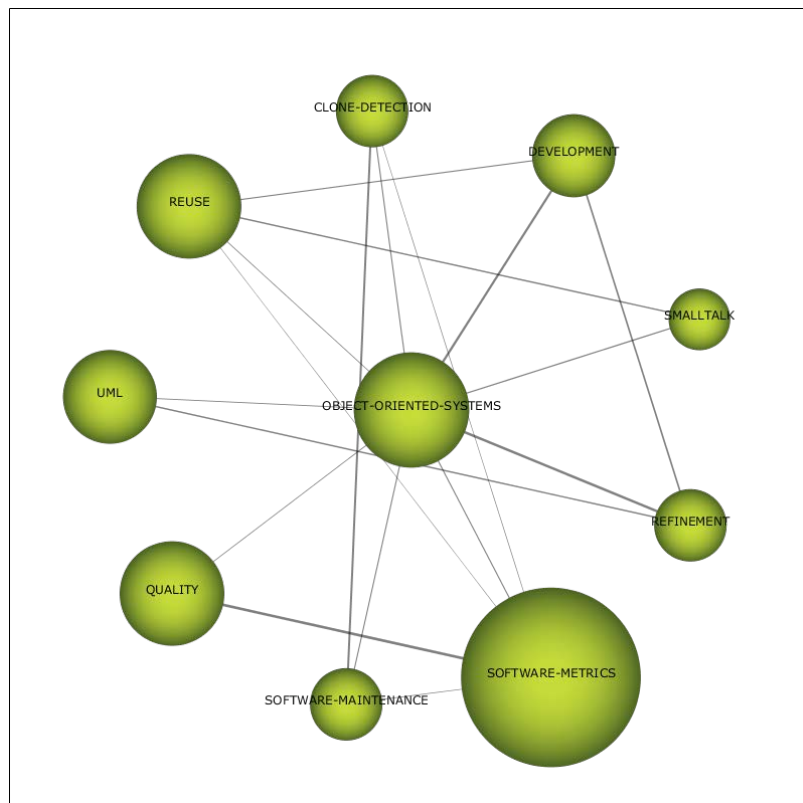


Figure 9: Thematic network *Object-Oriented-Systems* - period 2005-2009.